Survival Modeling of Cirrhosis Patients Using Applied Statistical Learning Methods

Course: Applied Statistical Learning (Stat 486)

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Abstract

Cirrhosis, a progressive chronic liver condition characterized by extensive scarring from long-term damage, poses significant challenges for patient survival prediction and risk stratification. The project evaluates whether machine learning–based survival models can improve prediction accuracy compared to the traditional Cox proportional hazards model, and to determine if these models can meaningfully stratify patients into clinically interpretable risk groups. Results demonstrate that LASSO-Cox achieved the highest predictive performance (C-index: 0.857), representing a marginal improvement over traditional Cox regression (C-index: 0.854).

Introduction

We use the cirrhosis dataset from Kaggle: https://www.kaggle.com/datasets/joebeachcapital/cirrhosis-patient-survival-prediction. The dataset used in this study originates from a Mayo Clinic investigation of primary biliary cirrhosis (PBC) conducted between 1974 and 1984, funded by the Mayo Clinic.

Accurate survival prediction is essential for clinical decision-making, including timing of liver transplantation, intensity of monitoring, and patient counseling. Traditional survival analysis relies on the Cox proportional hazards model, but machine learning methods offer potential advantages through automated feature selection, nonlinear relationships, and flexible interactions.

Research Objectives:

- Identify clinical variables that significantly affect mortality risk

- Compare predictive performance of ML survival models against traditional Cox regression
- Assess the impact of competing events (liver transplantation) on survival prediction
- Evaluate whether models can stratify patients into clinically meaningful risk groups

Dataset & Methods

a. Dataset & Data Processing

This dataset is from a Mayo Clinic study on primary biliary cirrhosis (PBC) conducted between 1974 and 1984, containing clinical data from 418 patients who participated in a randomized placebo-controlled trial of D-penicillamine. The dataset includes 17 clinical features encompassing demographic variables (age, sex), treatment assignment, clinical symptoms (ascites, hepatomegaly, spiders, edema), laboratory biomarkers, and histologic disease stage. Patient outcomes are categorized as death (D), censored alive (C), or censored due to liver transplantation (CL), with survival time measured in days from registration to the earlier of death, transplantation, or study analysis in July 1986. This longitudinal dataset provides a valuable resource for developing and comparing survival prediction models in liver cirrhosis research.

Data Processing:

- Missing data: Complete case analysis
- Train-test split: 70% training (n=292), 30% testing (n=126)
- Competing risk treatment: Liver Transplant coded as censored for primary analysis

b. Modeling Methods

- **Cox Proportional Hazards Model**

The Cox proportional hazards model serves as our basic approach.

- **LASSO-Cox Regression**

LASSO extends Cox regression by adding an L1 penalty term for automatic feature selection. The tuning parameter λ was selected via 10-fold cross-validation to minimize partial likelihood deviance.

- **Random Survival Forests**

RSF extends random forests to survival analysis by building an ensemble of survival trees. Each tree is grown on a bootstrap sample using random subsets of predictors. The method can capture complex nonlinear relationships without explicit specification.

- **Gradient Boosted Survival Models**

Gradient boosting builds the model step by step by adding one component at a time. Trees are added sequentially, each focusing on residuals from previous iterations. We used 10-fold cross-validation to determine the optimal number of trees.

Results & Outputs

a. Exploratory Data Analysis

The dataset contains 418 patients with a median age of 51 years (range: 26-78).

Approximately 35.4% died during follow-up, 26.1% were censored alive, and 38.5% underwent liver transplantation. Median follow-up was 1730 days (4.7 years).

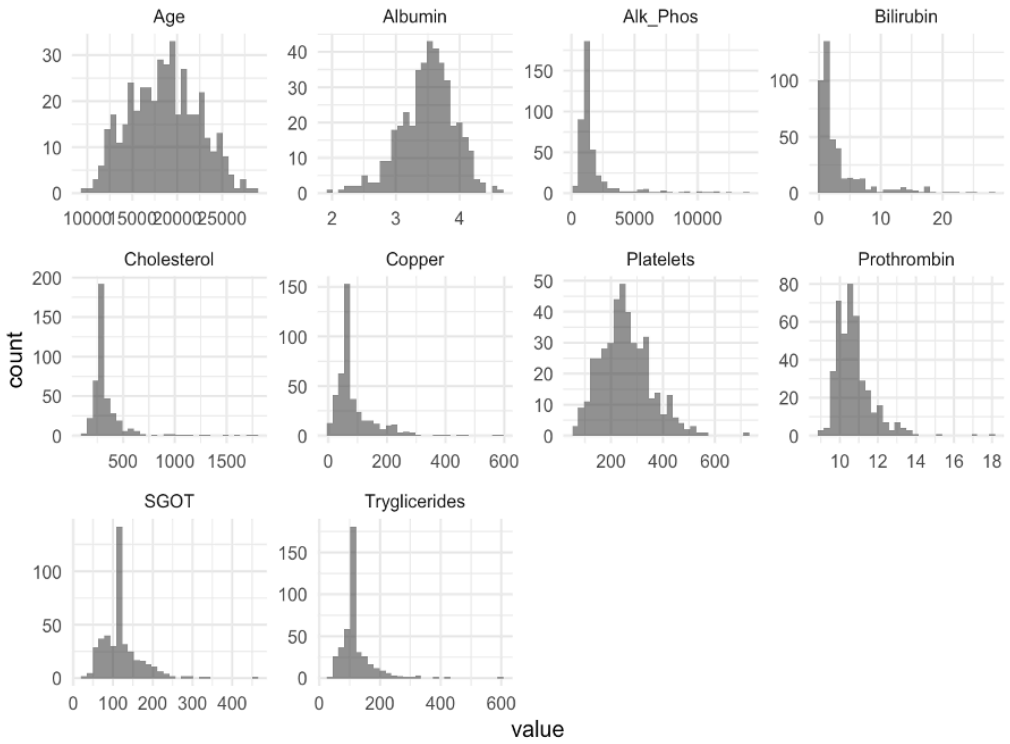


Figure 1: Distribution of key demographic and clinical variables

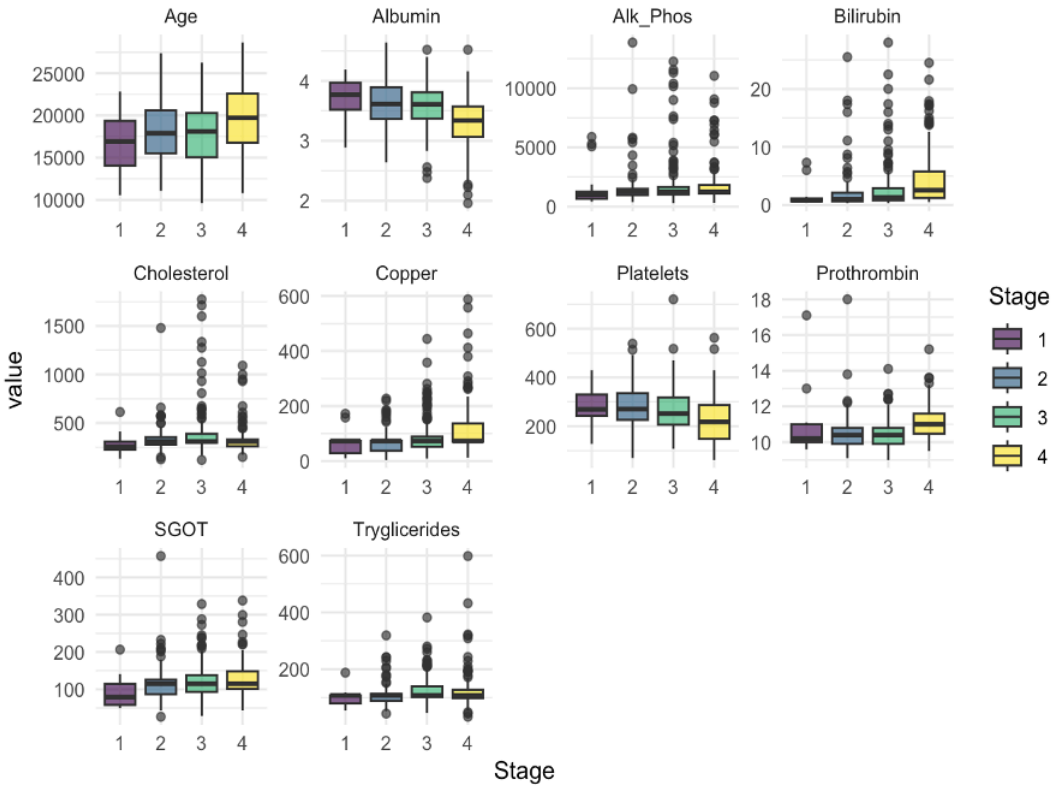


Figure 2: Laboratory biomarker distributions by survival status

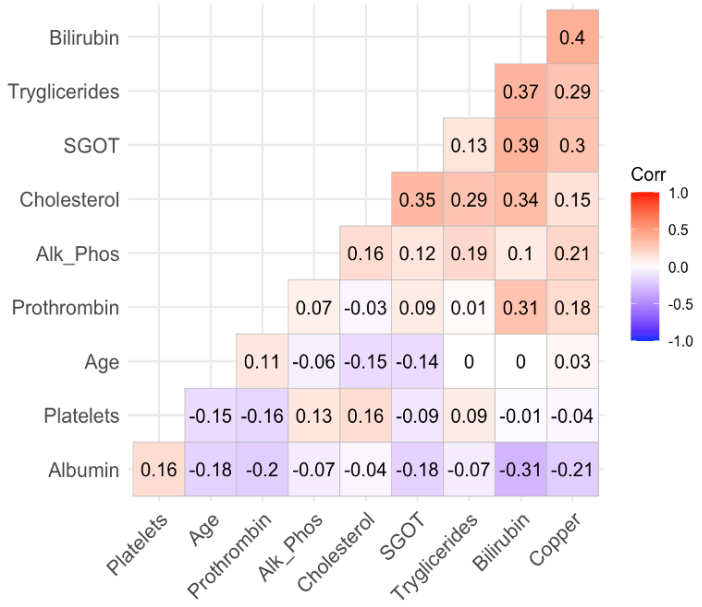
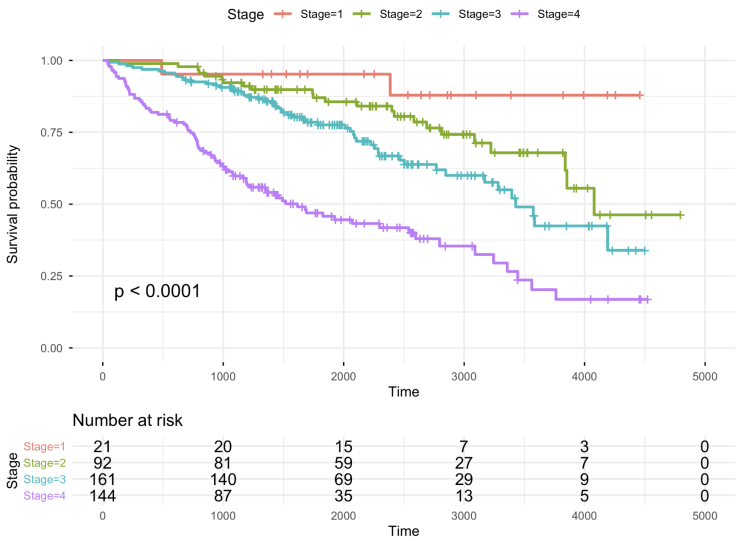
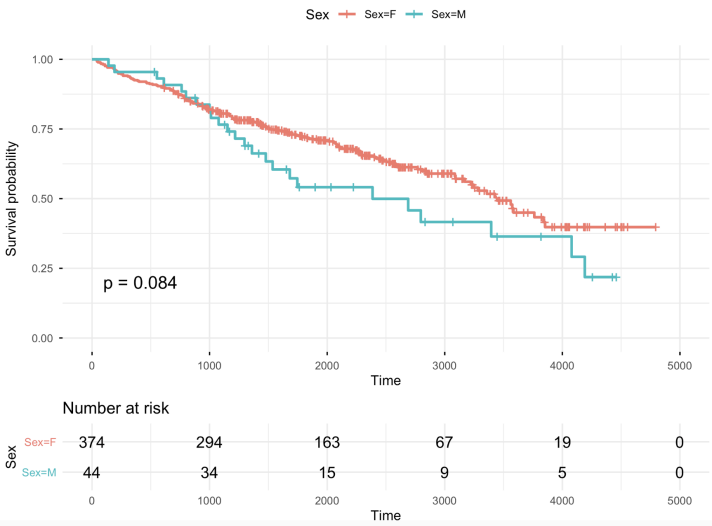


Figure 3: Correlation matrix of continuous predictors

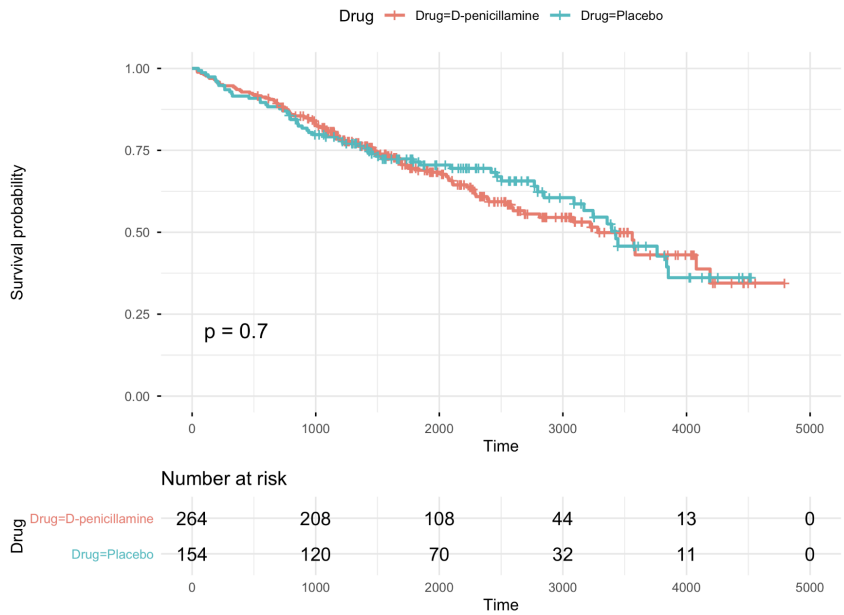
b. Kaplan-Meier Survival Analysis

Kaplan–Meier curves provided a basic description of overall survival. Patients with clinical features such as Ascites had much lower survival over time. These nonparametric estimates served as a reference for evaluating model-based risk stratification and demonstrated the clinical relevance of categorical disease severity indicators.





See the figures above. On the left is the Kaplan-Meier Curves by Disease Stage, $p < 0.0001$ implies that it's highly significant. On the right is the Kaplan-Meier Curves by Sex, which compares Female (F) with Male (M). The trend suggests that males have worse survival than females, but not statistically significant ($p = 0.084 > 0.05$).



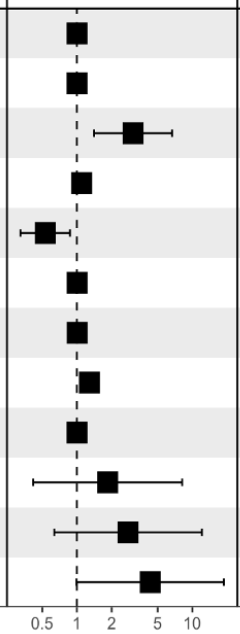
This figure above is the Kaplan-Meier Curves by Drug Treatment, which compares D-penicillamine with Placebo, implying that D-penicillamine does NOT significantly improve survival compared to placebo in this study.

c. Cox Proportional Hazards Model

The Cox model showed that having Ascites greatly increases the risk of death, and people with Ascites were about 3 times more likely to die than those without (HR=3.06). Albumin emerged as a protective factor, with higher levels associated with greatly reduced hazard. Laboratory biomarkers such as Bilirubin and Prothrombin time were highly significant, while disease stage lost significance once continuous biomarkers were included, suggesting that these markers capture underlying disease severity more precisely.

Below, a forest plot of hazard ratios shows the relative magnitude and direction of covariate effects, demonstrating the interpretability of the traditional Cox Model.

Variable		N	Hazard ratio	p
Age		292	1.00 (1.00, 1.00)	<0.001
Ascites	N	280	Reference	
	Y	12	3.06 (1.41, 6.66)	0.005
Bilirubin		292	1.10 (1.06, 1.14)	<0.001
Albumin		292	0.53 (0.32, 0.87)	0.012
Copper		292	1.00 (1.00, 1.01)	0.009
SGOT		292	1.00 (1.00, 1.01)	0.013
Prothrombin		292	1.28 (1.10, 1.49)	0.001
Stage	1	16	Reference	
	2	69	1.85 (0.42, 8.15)	0.418
	3	111	2.78 (0.64, 12.11)	0.174
	4	96	4.31 (0.99, 18.72)	0.051

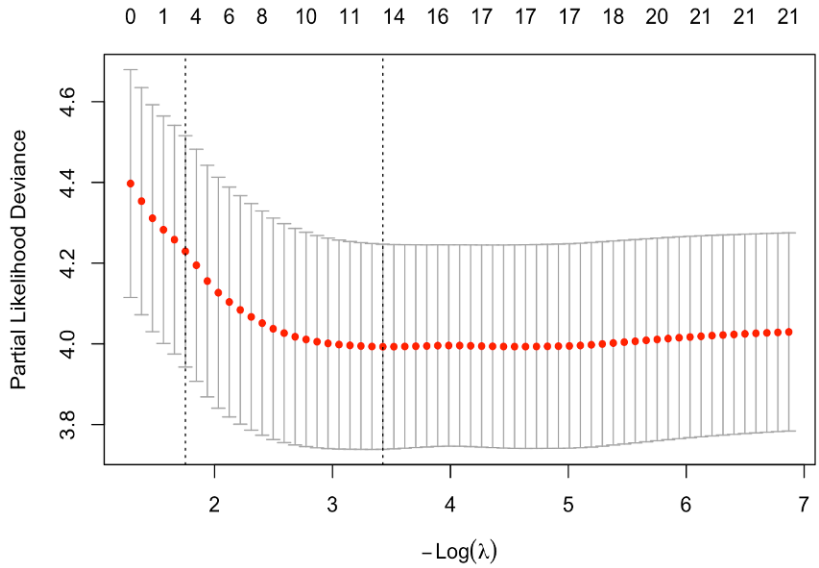


Key Findings:

- Ascites (HR=3.06, 95% CI: 1.41-6.66, p=0.005): Strongest predictor
- Albumin (HR=0.53, 95% CI: 0.32-0.87, p=0.012): Only protective factor
- Bilirubin (HR=1.10, 95% CI: 1.06-1.14, p<0.001): Highly significant cholestasis marker
- Prothrombin (HR=1.28, 95% CI: 1.10-1.49, p=0.001): Coagulation deficiency marker
- Disease stage became non-significant after controlling for continuous biomarkers

d. LASSO-Cox Model

The LASSO Cox model selected a subset of predictors. It retains key clinical variables while shrinking irrelevant coefficients to zero. Cross-validation identified an optimal regularization parameter corresponding to 13 non-zero coefficients. This model achieved the highest test-set C-index (0.8574), indicating a modest improvement over the traditional Cox model.



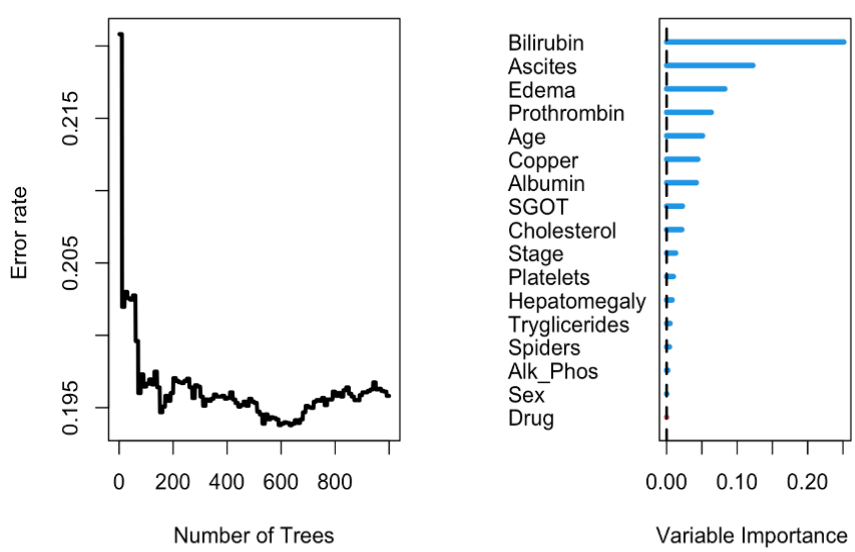

```
lasso_coefs <- coef(lasso_fit, s = cv_lasso$lambda.min)
print(lasso_coefs[lasso_coefs[, 1] != 0, , drop = FALSE])
```

```
## 13 x 1 sparse Matrix of class "dgCMatrix"
##                            1
## Age            7.732914e-05
## AscitesY       1.012036e+00
## HepatomegalyY  1.247299e-01
## SpidersY       1.379651e-01
## EdemaS         2.132034e-01
## Bilirubin      8.762915e-02
## Albumin       -5.127180e-01
## Copper         3.113760e-03
## Alk_Phos      -1.014358e-05
## SGOT           3.110159e-03
## Prothrombin    1.455219e-01
## Stage2        -4.595543e-02
## Stage4         4.279440e-01
```

Figure: LASSO cross-validation for regularization parameter selection

e. Random Survival Forests

RSF Model achieved C-index of 0.847. The RSF model demonstrated stable convergence with around 200 trees. Variable importance plot shows that Bilirubin is the dominant predictor, followed by Ascites, Edema, Prothrombin, and age. Variables such as drug assignment and sex showed negligible importance.

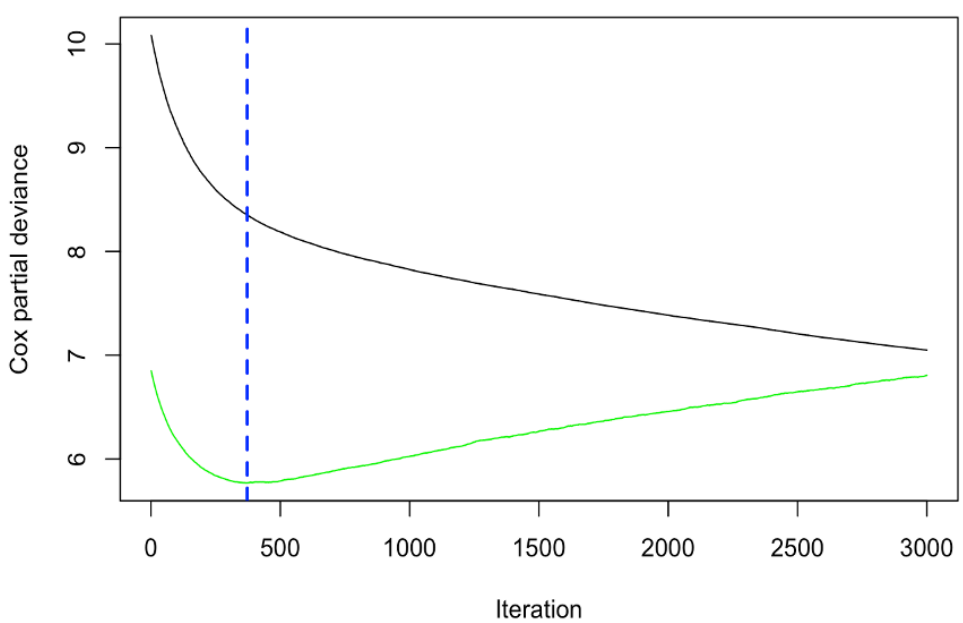


The left figure above is the OOB error rate versus number of trees, which converges rapidly in the first 100 trees and then becomes stable around 0.195 (80.5% accuracy) after 200 trees. The right figure displays variable importance rankings.

```
##
##                  Importance   Relative Imp
## Bilirubin            0.2512         1.0000
## Ascites              0.1223         0.4868
## Edema                0.0826         0.3287
## Prothrombin          0.0634         0.2522
## Age                  0.0511         0.2033
## Copper               0.0443         0.1764
## Albumin              0.0422         0.1680
## SGOT                 0.0225         0.0896
## Cholesterol          0.0216         0.0860
## Stage                0.0129         0.0512
## Platelets            0.0096         0.0384
## Hepatomegaly         0.0079         0.0316
## Tryglicerides        0.0051         0.0202
## Spiders              0.0041         0.0164
## Alk_Phos             0.0019         0.0074
## Sex                  0.0002         0.0009
## Drug                 0.0000        -0.0001
```

f. Gradient-boosted survival models/Cox

Cross-validation curves for the gradient-boosted survival model showed rapid decreases in error during early iterations, reaching a minimum at around 500 trees before slightly increasing, indicating overfitting beyond this point. The boosted model achieved competitive predictive performance but did not outperform the LASSO Cox model.



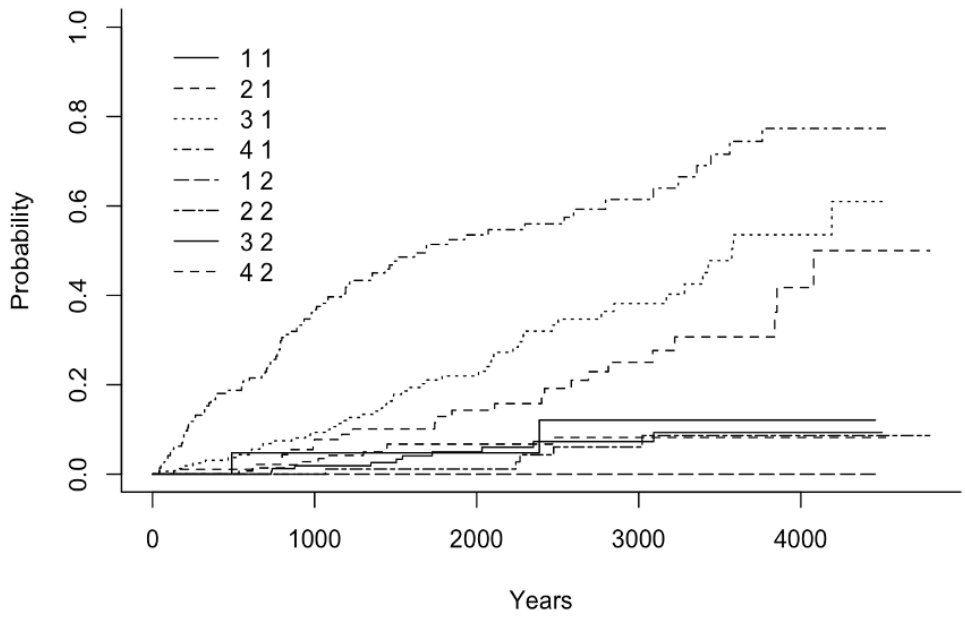
The plot above is the Cox Partial Deviance by Iteration (Gradient Boosting). The blue dashed line is the Optimal stopping point (minimum test deviance). Black curve is the Training set deviance. Green curve is the Test set deviance.

g. Competing Risks (Fine-Gray)

By treating liver transplantation as a competing event, the Fine-Gray model provided insight into cause-specific risk. Results suggested that key biomarkers affected both mortality and transplantation risk, indicating the importance of accounting for competing risks in cirrhosis prognosis.

Fine-Gray regression results：

```
## Competing Risks Regression
##
## Call:
## crr(ftime = ftime, fstatus = fstatus, cov1 = cov_for_cr, failcode = 1,
##     cencode = 0)
##
##                  coef exp(coef) se(coef)     z p-value
## Age          0.000096     1.000 2.42e-05  3.97 7.2e-05
## SexM         0.349540     1.418 2.62e-01  1.34 1.8e-01
## Bilirubin    0.124255     1.132 1.32e-02  9.39 0.0e+00
## Albumin     -0.781032     0.458 2.04e-01 -3.82 1.3e-04
## Prothrombin  0.266502     1.305 7.73e-02  3.45 5.6e-04
## Stage2       0.886746     2.427 7.99e-01  1.11 2.7e-01
## Stage3       1.389673     4.014 7.82e-01  1.78 7.6e-02
## Stage4       1.794303     6.015 7.80e-01  2.30 2.1e-02
##
##             exp(coef) exp(-coef)  2.5%  97.5%
## Age             1.000      1.000 1.000  1.000
## SexM            1.418      0.705 0.849  2.369
## Bilirubin       1.132      0.883 1.103  1.162
## Albumin         0.458      2.184 0.307  0.683
## Prothrombin     1.305      0.766 1.122  1.519
## Stage2          2.427      0.412 0.507 11.625
## Stage3          4.014      0.249 0.867 18.581
## Stage4          6.015      0.166 1.304 27.754
##
## Num. cases = 418
## Pseudo Log-likelihood = -784
## Pseudo likelihood ratio test = 195  on 8 df,
```



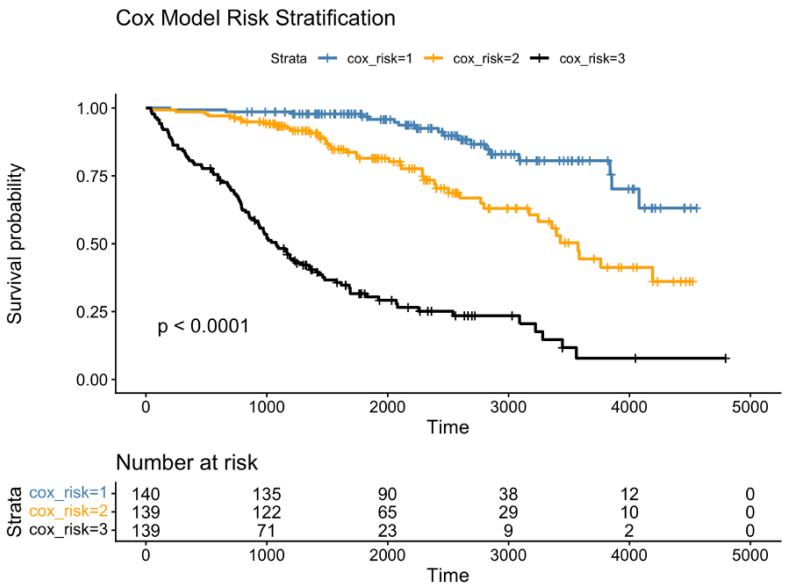
Interpretation of the plot above:

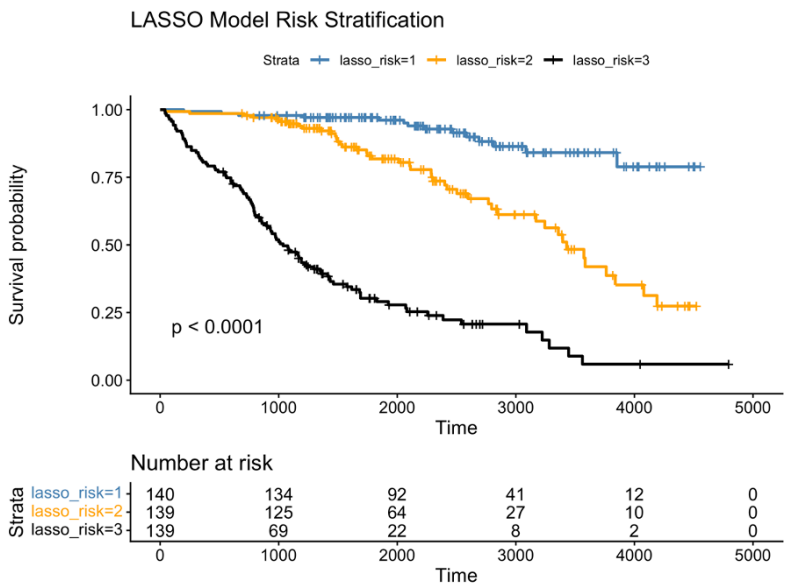
Event 1 is Death (Mortality), the primary outcome of interest. Curves: 1 1, 2 1, 3 1, 4 1 represent death rates across four risk groups.

Event 2: Liver Transplantation, which is our competing event that removes patients from the risk of death. Curves: 1 2, 2 2, 3 2, 4 2 show transplant rates across the same risk groups.

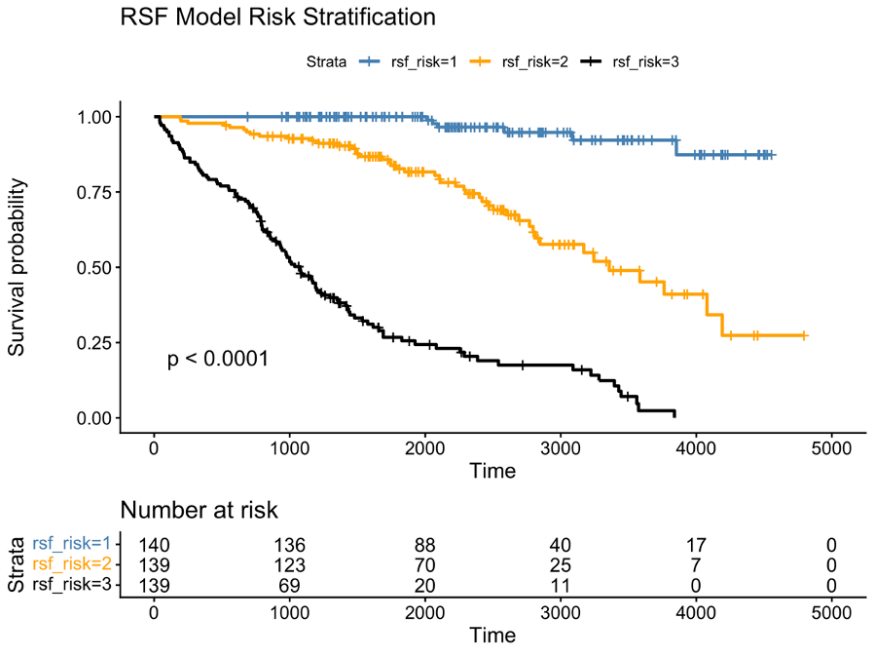
h. Risk Stratification (Cox / LASSO / RSF)

Using predicted risk scores, patients were stratified into high, medium, and low-risk groups.





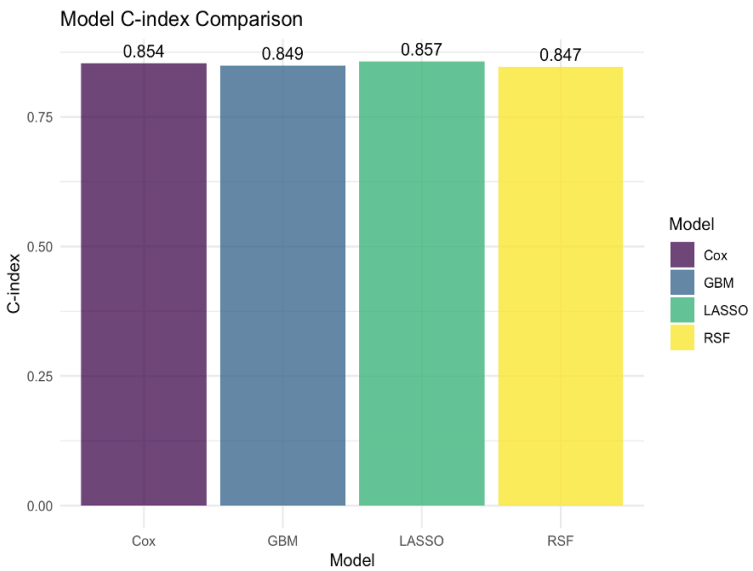
Left: Kaplan-Meier curves by Cox model risk groups *Right: Kaplan-Meier curves by LASSO model risk groups*

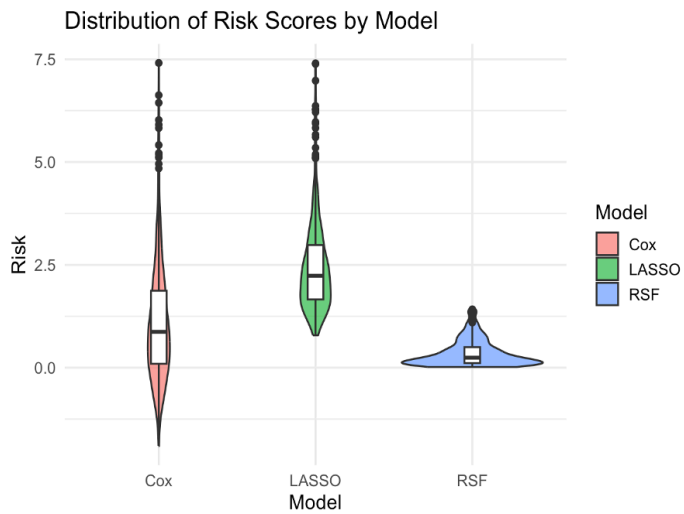


Kaplan-Meier curves by RFS model risk groups

i. Model Performance Summary

Across all models, C-index values indicated that the LASSO Cox model achieved the best balance between predictive accuracy and interpretability. It offered small but consistent improvements over the traditional Cox model. RSF and gradient boosting helped confirm nonlinear effects and identify important variables.





On the left is the C-index bar chart showing that all four models (Cox, GBM, LASSO, RSF) achieve similar excellent discrimination (C-index 0.847-0.857). LASSO performs best at a C-index of 0.857. On the right is the distribution of risk scores by model, which are violin plots. Cox and LASSO have similar distributions while RSF generates risk scores on a compressed scale near zero (range 0-1.5). It shows that despite comparable C-indices, different models produce risk scores on different scales.

Conclusion

The results indicate that LASSO-Cox (C-index: 0.857) Model, achieves marginal improvement over traditional Cox regression while providing automatic feature selection. All models identify liver function biomarkers (Bilirubin, Albumin, Prothrombin) as key predictors and successfully stratify patients into risk groups ($p < 0.001$).

These findings support the use of regularized statistical learning for survival prediction and emphasize that model interpretability remains crucial for clinical implementation. For clinical use, we recommend LASSO-Cox for its balanced performance, while traditional Cox remains a strong choice.

Acknowledgements

We acknowledge the Mayo Clinic for the original primary biliary cirrhosis study and the Kaggle community for curating and sharing the dataset used in this project.

Appendix

We will also submit the Cirrhosis. Html file.